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**The workflow of the EPO-RSAC algorithm**

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**Step 1: Training initialization**

1. Set confidence level  $\beta$  and learning rates  $\omega$
2. Initialize policy parameters  $\theta$ , critic parameters  $\varphi_1, \varphi_2, \mu, \eta$ , Lagrange multipliers  $\alpha, \kappa$
3. Create target-network copies  $\bar{\varphi}_1, \bar{\varphi}_2, \bar{\mu}, \bar{\eta}$
4. Initialize the experience buffer  $\mathcal{D} \leftarrow \emptyset$

**Step 2: Episode Loop**

Repeat until convergence:

**Step 2.1: Environment Interaction**

Reset the environment to an initial state  $s_0$

For each time step  $t$  in the episode:

1. Action Selection: Sample  $a_t \sim \pi_\theta(a_t|s_t)$
2. Reward and Cost Calculation:  
Execute action within upper- and lower-level to obtain  $r_t$  and  $c_t$
3. State Transition:  $s_{t+1} = \mathcal{P}(s_{t+1}|s_t, a_t)$
4. Experience Storage: Append  $(s_t, a_t, r_t, c_t, s_{t+1}, done)$  to  $\mathcal{D}$

**Step 2.2: Gradient Steps****Step 2.2.1: Mini-batch Sampling**

Sample a random batch of transitions from  $\mathcal{D}$

**Step 2.2.2: Importance Ratio Computation**

Compute the security-critic function  $\Gamma_{\pi_\theta}$  as per Eq. (31).

**Step 2.2.3: Critic Updates**

1. Reward critics:  $\varphi_i \leftarrow \varphi_i - \omega \nabla_{\varphi} J_r(\varphi_i)$  for  $i \in \{1, 2\}$  by (35)
2. Cost critic:  $\mu \leftarrow \mu - \omega \nabla_{\mu} J_c(\mu)$  by (36)
3. Cost-variance critic:  $\eta \leftarrow \eta - \omega \nabla_{\eta} J_v(\eta)$  by (37)

**Step 2.2.4: Actor Update**

Primal policy update:  $\theta \leftarrow \theta - \omega \nabla_{\theta} \mathcal{L}(\pi_\theta, \alpha, \kappa)$  by (39)

**Step 2.2.5: Dual Variable Update**

1. Lagrange multipliers- $\alpha$ :  $\alpha \leftarrow \alpha - \omega \nabla_{\alpha} \mathcal{L}(\pi_\theta, \alpha, \kappa)$  by (40)
2. Lagrange multipliers- $\kappa$ :  $\kappa \leftarrow \kappa - \omega \nabla_{\kappa} \mathcal{L}(\pi_\theta, \alpha, \kappa)$  by (41)

**Step 2.2.6: Target-Network Soft Update**

1. Reward critics:  $\bar{\varphi}_i \leftarrow \omega_{soft} \varphi_i + (1 - \omega_{soft}) \bar{\varphi}_i$  for  $i \in \{1, 2\}$
2. Cost critic:  $\bar{\mu} \leftarrow \omega_{soft} \mu + (1 - \omega_{soft}) \bar{\mu}$
3. Cost-variance critic:  $\bar{\eta} \leftarrow \omega_{soft} \eta + (1 - \omega_{soft}) \bar{\eta}$

**Step 3: Termination and Output**

Upon meeting convergence criteria, output the final policy parameters:

$$\theta^* = \arg \min_{\alpha \geq 0} \min_{\kappa \geq 0} \max_{\theta} \mathcal{L}(\pi_\theta, \alpha, \kappa)$$

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**Algorithm 1** Robust SAC With Embedded Physical Optimization (EPO-RSAC)

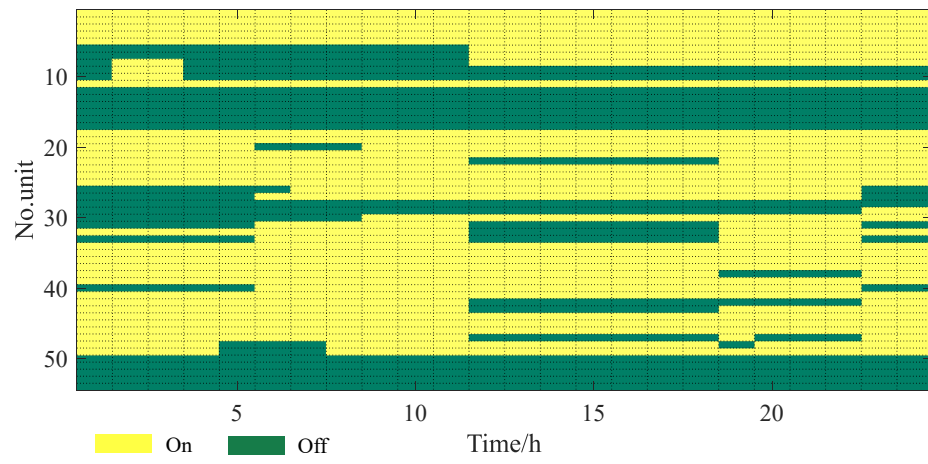
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1: Initialize:  $\beta=0.95$ ,  $\omega$ ,  $\varphi$ ,  $\mu$ ,  $\eta$ ,  $\theta$ ,  $\alpha$ ,  $\kappa$ ,  $\bar{\varphi}, \bar{\mu}, \bar{\eta}$ ,  $\mathcal{D} \leftarrow \emptyset$ 
2: foreach episode do
3:   foreach environment step do
4:     Decision-making  $a_t \sim \pi_\theta(a_t|s_t)$ 
5:     Calculating  $r_t, c_t$  by upper- and lower-level
6:     Executing state transition  $s_{t+1} = \mathcal{P}(s_{t+1}|s_t, a_t)$ 
7:      $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s_t, a_t, r_t, c_t, s_{t+1}, done)\}$ 
8:   end
9:   foreach gradient step do
10:    Mini-batch Sampling from  $\mathcal{D}$ 
11:    Computing  $\Gamma_{\pi^\theta}$  based on (31)
12:    # update critic network for RCN and SCN
13:     $\varphi_i \leftarrow \varphi_i - \omega \nabla_{\varphi} J_i(\varphi_i)$  for  $i \in \{1, 2\}$  by (35)
14:     $\mu \leftarrow \mu - \omega \nabla_{\mu} J_c(\mu)$  by (36)
15:     $\eta \leftarrow \eta - \omega \nabla_{\eta} J_v(\eta)$  by (37)
16:    # primal policy update for actor network
17:     $\theta \leftarrow \theta - \omega \nabla_{\theta} \mathcal{L}(\pi_\theta, \alpha, \kappa)$  by (39)
18:    # dual LM update for  $\alpha, \kappa$ 
19:     $\alpha \leftarrow \alpha - \omega \nabla_{\alpha} \mathcal{L}(\pi_\theta, \alpha, \kappa)$  by (40)
20:     $\kappa \leftarrow \kappa - \omega \nabla_{\kappa} \mathcal{L}(\pi_\theta, \alpha, \kappa)$  by (41)
21:    # Soft update for target networks with  $\omega_{soft}$ 
22:     $\bar{\varphi}_i \leftarrow \omega_{soft} \varphi_i + (1 - \omega_{soft}) \bar{\varphi}_i$  for  $i \in \{1, 2\}$ 
23:     $\bar{\mu} \leftarrow \omega_{soft} \mu + (1 - \omega_{soft}) \bar{\mu}$ 
24:     $\bar{\eta} \leftarrow \omega_{soft} \eta + (1 - \omega_{soft}) \bar{\eta}$ 
25:   end
26: end
27: Output:  $\theta^* = \arg \min_{\alpha \geq 0} \min_{\kappa \geq 0} \max_{\theta} \mathcal{L}(\pi_\theta, \alpha, \kappa)$ 
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TABLE I EPO-RSAC NEURAL NETWORK ARCHITECTURE AND PARAMETERS FOR IEEE-118 SYSTEM

| Variable scope      | name          | Parameter           | Configuration                                                         |
|---------------------|---------------|---------------------|-----------------------------------------------------------------------|
| Main                | Actor         | Number of neurons   | (118, 64, 256, 256, 64, 120(mu))<br>(118, 64, 256, 256, 64, 120(std)) |
|                     |               | Activation function | ReLU                                                                  |
|                     |               | Optimizer           | Adam                                                                  |
|                     |               | Learning rate       | 1~4000 (1e-3), 4001~10000 (1e-4)                                      |
|                     | Critic_qr1    | Number of neurons   | (238, 64, 256, 256, 64, 1)                                            |
|                     |               | Activation function | ReLU                                                                  |
|                     |               | Optimizer           | Adam                                                                  |
|                     |               | Learning rate       | 1~4000 (1e-3), 4001~10000 (1e-4)                                      |
|                     | Critic_qr2    | Number of neurons   | (238, 64, 256, 256, 64, 1)                                            |
|                     |               | Activation function | ReLU                                                                  |
|                     |               | Optimizer           | Adam                                                                  |
|                     |               | Learning rate       | 1~4000 (1e-3), 4001~10000 (1e-4)                                      |
|                     | Critic_qc     | Number of neurons   | (238, 64, 256, 256, 64, 1)                                            |
|                     |               | Activation function | ReLU                                                                  |
|                     |               | Optimizer           | Adam                                                                  |
|                     |               | Learning rate       | 1~4000 (1e-3), 4001~10000 (1e-4)                                      |
|                     | Critic_var    | Number of neurons   | (238, 64, 256, 256, 64, 1)                                            |
|                     |               | Activation function | ReLU                                                                  |
|                     |               | Optimizer           | Adam                                                                  |
|                     |               | Learning rate       | 1~4000 (1e-3), 4001~10000 (1e-4)                                      |
|                     | Replay buffer | Size of replay      | 24000                                                                 |
|                     |               | Batch size          | 256                                                                   |
| Target              | Actor         | Number of neurons   | (118, 64, 256, 256, 64, 120(mu))<br>(118, 64, 256, 256, 64, 120(std)) |
|                     |               | Soft update rate    | 0.995                                                                 |
|                     | Critic_qr1    | Number of neurons   | (238, 64, 256, 256, 64, 1)                                            |
|                     |               | Soft update rate    | 0.995                                                                 |
|                     | Critic_qr2    | Number of neurons   | (238, 64, 256, 256, 64, 1)                                            |
|                     |               | Soft update rate    | 0.995                                                                 |
|                     | Critic_qc     | Number of neurons   | (238, 64, 256, 256, 64, 1)                                            |
|                     |               | Soft update rate    | 0.995                                                                 |
|                     | Critic_var    | Number of neurons   | (238, 64, 256, 256, 64, 1)                                            |
|                     |               | Soft update rate    | 0.995                                                                 |
| Entropy temperature | Soft $\alpha$ | Size of weight      | 1                                                                     |
|                     |               | initializer         | 0                                                                     |
|                     |               | Optimizer           | Adam                                                                  |
|                     |               | Learning rate       | 1~4000 (1e-3), 4001~10000 (1e-4)                                      |
| Cost weight         | Soft $\kappa$ | Size of weight      | 1                                                                     |
|                     |               | initializer         | 0                                                                     |
|                     |               | Optimizer           | Adam                                                                  |
|                     |               | Learning rate       | 1~4000 (1e-3), 4001~10000 (1e-4)                                      |



**Fig. 1:** Commitment decisions formulated by EPO-RSAC for IEEE118 bus system